Cairo University CHS - team 8

Faculty of Engineering

Computer Engineering Department CMPS202

Computer Engineering Department

CMPS202

Introduction to Database Systems MochaBase

Database Schema Report

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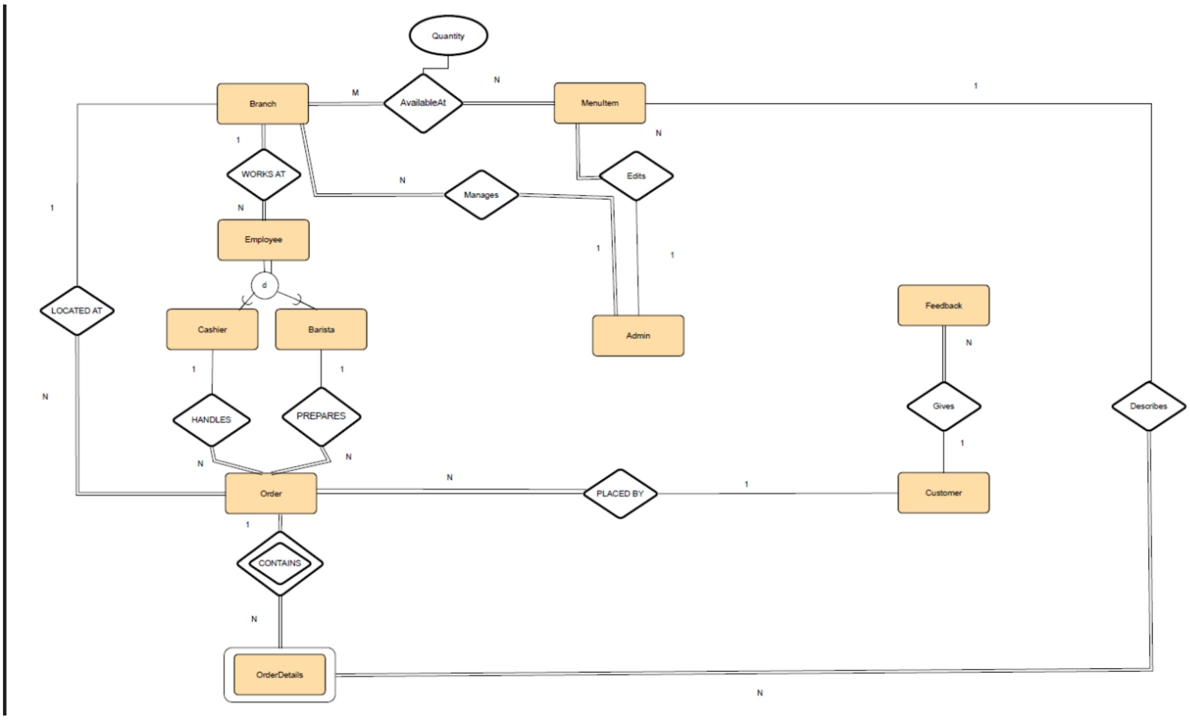
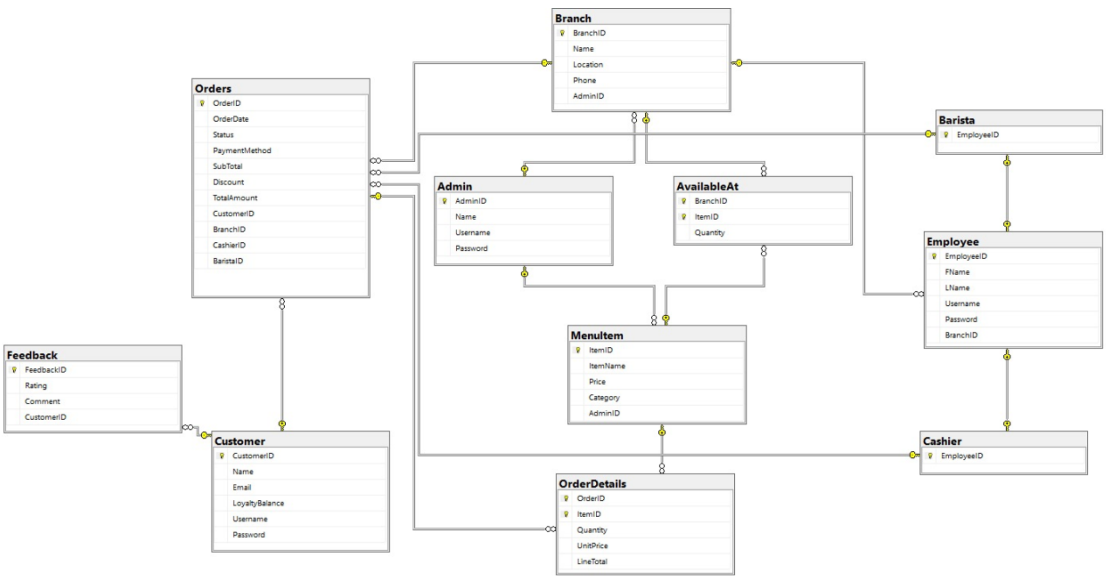
Updated ER digram :



Database Schema:



Database constraints:

Database Constraints Report

Mocha Cafe Management System

Executive Summary

This report documents all constraints implemented in the Mocha database to ensure data integrity, enforce business rules, and maintain referential integrity across all tables. The database consists of 11 tables with various constraint types including primary keys, foreign keys, unique constraints, check constraints, and default values.

1. Primary Key Constraints

Primary keys uniquely identify each record in a table and automatically create a clustered index.

1.1 Single-Column Primary Keys

Table	Column	Type	Description
Admin	AdminID	INT IDENTITY(1,1)	Auto-incrementing unique identifier for administrators
Customer	CustomerID	INT IDENTITY(1,1)	Auto-incrementing unique identifier for customers
Branch	BranchID	INT IDENTITY(1,1)	Auto-incrementing unique identifier for branches
MenuIt	ItemID	INT IDENTITY(1,1)	Auto-incrementing unique identifier for menu items
Employee	EmployeeID	INT IDENTITY(1,1)	Auto-incrementing unique identifier for employees
Cashier	EmployeeID	INT	References Employee table
Barista	EmployeeID	INT	References Employee table
Orders	OrderID	INT IDENTITY(1,1)	Auto-incrementing unique identifier for orders
Feedback	FeedbackID	INT IDENTITY(1,1)	Auto-incrementing unique identifier for feedback

1.2 Composite Primary Keys

Table	Columns	Purpose
AvailableAt	(BranchID, ItemID)	Ensures unique inventory record per branch-item combination
OrderDetails	(OrderID, ItemID)	Ensures unique line item per order-item combination

2. Foreign Key Constraints

Foreign keys establish relationships between tables and enforce referential integrity.

2.1 Admin-Related Relationships

Child Table	Foreign Key Column	Parent Table	Parent Column	Purpose
Branch	AdminID	Admin	AdminID	Tracks which admin manages each branch
MenuItem	AdminID	Admin	AdminID	Tracks which admin created each menu item

2.2 Branch-Related Relationships

Child Table	Foreign Key Column	Parent Table	Parent Column	Cascade Delete
Employee	BranchID	Branch	BranchID	No
Orders	BranchID	Branch	BranchID	No
AvailableAt	BranchID	Branch	BranchID	Yes

2.3 Employee-Related Relationships

Child Table	Foreign Key Column	Parent Table	Parent Column	Cascade Delete
Cashier	EmployeeID	Employee	EmployeeID	Yes
Barista	EmployeeID	Employee	EmployeeID	Yes
Orders	CashierID	Cashier	EmployeeID	No
Orders	BaristaID	Barista	EmployeeID	No

Note: The Cashier and Barista tables implement inheritance from Employee table.

2.4 Menu & Inventory Relationships

Child Table	Foreign Key Column	Parent Table	Parent Column	Cascade Delete
AvailableAt	ItemID	MenuItem	ItemID	Yes
OrderDetails	ItemID	MenuItem	ItemID	No

2.5 Order-Related Relationships

Child Table	Foreign Key Column	Parent Table	Parent Column	Cascade Delete
Orders	CustomerID	Customer	CustomerID	No
OrderDetails	OrderID	Orders	OrderID	Yes

2.6 Customer-Related Relationships

Child Table	Foreign Key Column	Parent Table	Parent Column	Cascade Delete
Feedback	CustomerID	Customer	CustomerID	Yes

3. Unique Constraints

Unique constraints ensure no duplicate values exist in specified columns.

Table	Column	Purpose
Admin	Username	Prevents duplicate admin usernames
Customer	Username	Prevents duplicate customer usernames
Employee	Username	Prevents duplicate employee usernames

Business Rule: All users (admins, customers, employees) must have unique usernames for authentication purposes.

4. Check Constraints

Check constraints enforce domain integrity by limiting values that can be entered.

4.1 Orders Table

Column	Constraint	Allowed Values

Status	CHECK (Status IN (...))	'Pending', 'Preparing', 'Ready', 'Completed'
PaymentMethod	CHECK (PaymentMethod IN (...))	'Cash', 'Card', 'Loyalty'

Purpose: Ensures order status follows the workflow and payment methods are valid.

4.2 OrderDetails Table

Column	Constraint	Rule
Quantity	CHECK (Quantity > 0)	Must be positive integer

Purpose: Prevents orders with zero or negative quantities.

4.3 Feedback Table

Column	Constraint	Allowed Values
Rating	CHECK (Rating BETWEEN 1 AND 5)	1, 2, 3, 4, or 5

Purpose: Ensures ratings follow a standard 5-star scale.

5. Default Constraints

Default constraints provide automatic values when no value is specified.

Table	Column	Default Value	Purpose
Customer	LoyaltyBalance	0	New customers start with zero loyalty points
AvailableAt	Quantity	0	New inventory items start with zero stock
Orders	OrderDate	CURRENT_TIMESTAMP	Automatically records order creation time
Orders	Discount	0.00	Orders have no discount unless specified

6. NOT NULL Constraints

NOT NULL constraints ensure required fields always have values.

6.1 User Authentication Tables

Admin Table:

- Name, Username, Password (all NOT NULL)

Customer Table:

- Name, Username, Password (all NOT NULL)

Employee Table:

- FName, LName, Username, Password, BranchID (all NOT NULL)

Purpose: Ensures all user accounts have complete authentication information.

6.2 Business Entity Tables

Branch Table:

- Name, Location, AdminID (all NOT NULL)

MenuItem Table:

- ItemName, Price, AdminID (all NOT NULL)

Purpose: Ensures core business data is always complete.

6.3 Transaction Tables

Orders Table:

- SubTotal, TotalAmount, CustomerID, BranchID, CashierID, BaristaID (all NOT NULL)

OrderDetails Table:

- Quantity, UnitPrice, LineTotal (all NOT NULL)

Purpose: Ensures financial transactions have complete information.

7. Cascade Delete Rules

Cascade delete automatically removes related records when a parent record is deleted.

7.1 Tables WITH Cascade Delete

Parent Table	Child Table	Rationale
Branch	AvailableAt	Inventory is branch-specific; remove when branch closes
MenuItem	AvailableAt	Inventory is item-specific; remove when item discontinued
Employee	Cashier	Role is employee-specific; remove when employee deleted
Employee	Barista	Role is employee-specific; remove when employee deleted
Orders	OrderDetails	Line items belong to order; remove when order deleted
Customer	Feedback	Feedback is customer-specific; remove for privacy

7.2 Tables WITHOUT Cascade Delete

Parent Table	Child Table	Rationale
Branch	Employee	Preserve employee records for audit trail
Branch	Orders	Preserve order history for financial records
Employee	Orders	Preserve order history (via Cashier/Barista)
MenuItem	OrderDetails	Preserve historical pricing data
Customer	Orders	Preserve order history for business analytics

Business Rule: Historical transaction data must be preserved for auditing and analytics purposes.

8. Constraint Validation Rules

8.1 Data Entry Validation

1. **Username Uniqueness:** System must check for existing usernames before account creation
2. **Order Status Flow:** Orders must progress through valid states only
3. **Quantity Validation:** All quantities must be positive integers
4. **Rating Validation:** Feedback ratings must be 1-5 stars only
5. **Payment Method:** Only accepted payment methods are allowed

8.2 Deletion Restrictions

1. **Cannot delete Branch if:**

   a. Branch has any orders (historical data preservation)
2. **Cannot delete Employee if:**
   a. Employee has processed any orders (via Cashier/Barista roles)
3. **Cannot delete MenuItem if:**
   a. Item appears in any order details (historical pricing)
4. **Cannot delete Customer if:**
   a. Customer has any orders (financial records)

9. Referential Integrity Summary

9.1 Strong Relationships (Mandatory)

- Every **Branch** must have an **Admin**
- Every **MenuItem** must have an **Admin**
- Every **Employee** must belong to a **Branch**
- Every **Order** must have a **Customer**, **Branch**, **Cashier**, and **Barista**
- Every **OrderDetails** record must reference a valid **Order** and **MenuItem**
- Every **Feedback** must reference a **Customer**

9.2 Weak Relationships (Optional)

- **None** - All relationships in this database are mandatory

10. Constraint Impact on Application Logic

10.1 Required Application Validations

1. **Before adding Branch or MenuItem:** Verify AdminID exists
2. **Before adding Employee:** Verify BranchID exists
3. **Before creating Order:** Verify CustomerID, BranchID, CashierID, BaristaID exist
4. **Before deleting Branch:** Check for related orders
5. **Before deleting Employee:** Check for order history

10.2 Error Handling Requirements

The application must handle these constraint violations:

- **Primary Key Violation:** Duplicate ID (should not occur with IDENTITY)
- **Foreign Key Violation:** Referenced parent record doesn't exist
- **Unique Constraint Violation:** Username already exists

- **Check Constraint Violation:** Invalid status, payment method, quantity, or rating
- **NOT NULL Violation:** Required field is empty

11. Database Integrity Benefits

11.1 Data Quality

- **Completeness:** NOT NULL constraints ensure required data is present
- **Accuracy:** CHECK constraints ensure valid values
- **Uniqueness:** PRIMARY KEY and UNIQUE constraints prevent duplicates
- **Consistency:** FOREIGN KEY constraints maintain relationships

11.2 Business Rule Enforcement

- **Order Workflow:** Status CHECK constraint enforces business process
- **Inventory Management:** Composite keys prevent duplicate inventory records
- **User Management:** UNIQUE usernames prevent account conflicts
- **Historical Data:** Selective cascade rules preserve audit trails

11.3 Security

- **Referential Integrity:** Prevents orphaned records
- **Data Privacy:** CASCADE DELETE on Feedback removes personal data with customer
- **Audit Trail:** Non-cascading deletes preserve transaction history
- # Conclusion

The Mocha database implements a robust constraint system that ensures:

- **Data Integrity:** All data is valid, complete, and consistent
- **Referential Integrity:** All relationships are properly maintained
- **Business Rules:** Application logic is enforced at database level
- **Historical Preservation:** Transaction data is protected from deletion
- **User Management:** Authentication data is properly validated